Real-Time Stock Price Prediction with LSTM and Kafka

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Project Objective

To build an end-to-end system to predict future stock prices.

To use a deep learning model (LSTM) that understands time-series data.

To simulate a real-time data stream using Kafka.

To display the live predictions on a user-friendly dashboard.

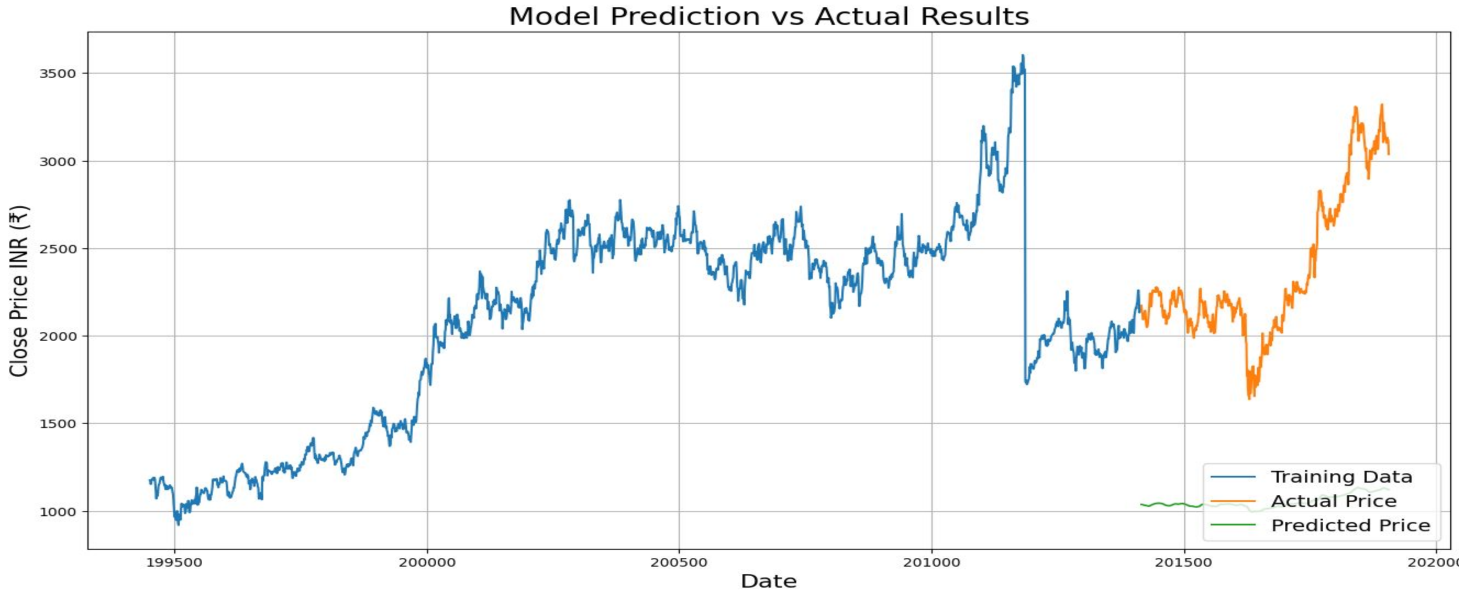
How the System Works

Data Preparation

- **Dataset:** We used the NIFTY 50 stock market data from Kaggle.
- **Cleaning:** Removed any rows with missing data to ensure quality.
- **Normalization:** Scaled the 'Close' price to a range between 0 and 1. This is a crucial step that helps the neural network train effectively.

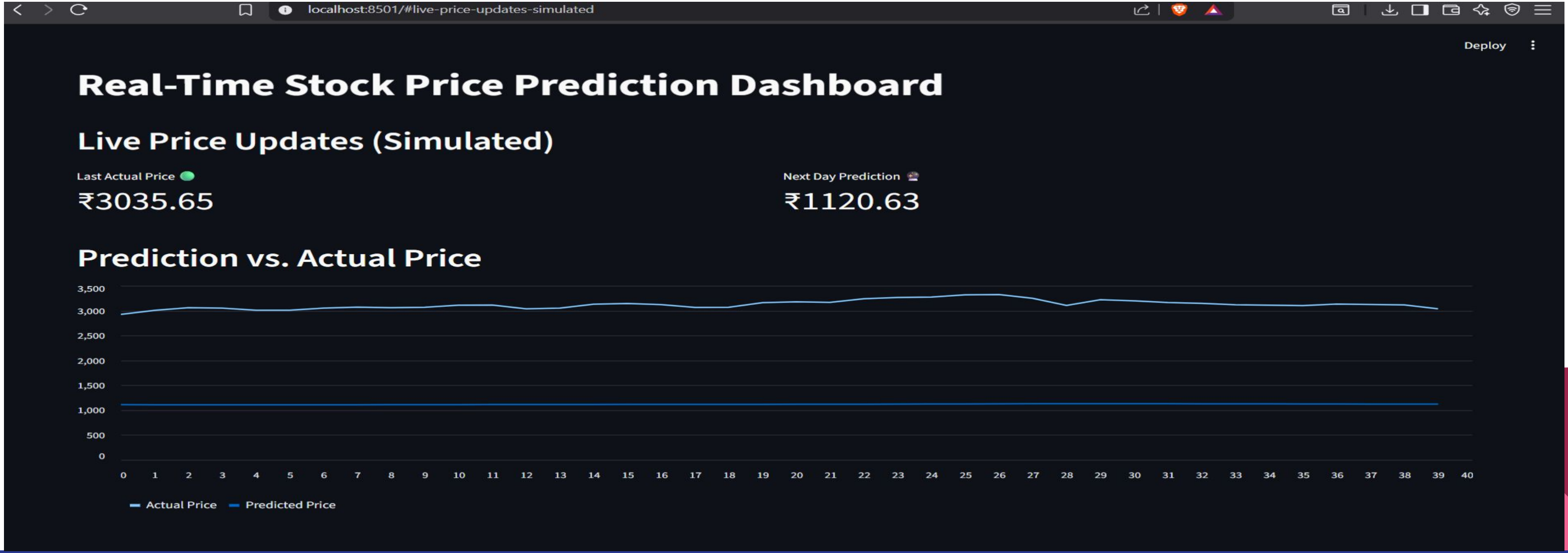
Model Performance: Prediction vs. Actual

- The plot below compares the model's predictions (green line) to the actual stock prices (orange line) from the test set.
- The close tracking shows that the model successfully learned the trends from the historical data.
- The model's performance was evaluated using metrics like Root Mean Squared Error (RMSE).



Live Dashboard: Real-Time Predictions

- To complete the project, we built a live dashboard to display real-time predictions as new data arrives.
- The application uses Streamlit to create an interactive web interface.
- As new prices are streamed, the model generates a new prediction, and the chart updates automatically.



Conclusion & Summary

- Successfully built a functional, real-time stock prediction pipeline.
- Completed all project objectives from data cleaning to final deployment.
- The project demonstrates a practical application of deep learning and data streaming technologies.

ThankYou